Lecture 9: Section 2.6

Chain rule

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The Idea Behind Chain Rule

Real-World Analogy

Think of a factory assembly line:

- Machine A takes raw materials and makes components
- Machine B takes components and makes final products
- The **rate** of final production depends on **both** machines

Example

- $f(x)$: Distance traveled (miles) as function of time (hours)
- $g(t)$: Fuel consumption (gallons) as function of distance (miles)
- $g(f(x))$: Fuel consumption as function of time
- Rate of fuel use = (Rate of fuel per mile) $\times$ (Rate of miles per hour)

Composition of Functions

Definition

If f and g are functions, the **composition** $g \circ f$ is defined as:

$$(g \circ f)(x) = g(f(x))$$

Example

Let $f(x) = x^2 + 1$ and $g(x) = \sqrt{x}$, then:

$$(g \circ f)(x) = g(f(x)) = g(x^2 + 1) = \sqrt{x^2 + 1}$$

Visualizing Composition

$$x \longrightarrow \boxed{f} \longrightarrow f(x) \longrightarrow \boxed{g} \longrightarrow g(f(x))$$

The Chain Rule Theorem

In Leibniz Notation

If $y = f(u)$ and $u = g(x)$, then:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Theorem (Chain Rule)

If g is differentiable at x and f is differentiable at $g(x)$, then the composite function $f \circ g$ is differentiable at x, and:

$$(f \circ g)'(x) = f'(g(x)) \cdot g'(x)$$

Why Chain Rule Makes Sense

Verbal Description

The derivative of a composition is:

(Derivative of outer function evaluated at inner function) $\times$ (Derivative of inner function)

Incremental Thinking

- If $u = g(x)$, then $\Delta u \approx g'(x)\Delta x$
- If $y = f(u)$, then $\Delta y \approx f'(u)\Delta u$
- So: $\Delta y \approx f'(u) \cdot g'(x)\Delta x$
- Thus: $\frac{dy}{dx} = f'(u) \cdot g'(x)$

Example

If cars travel at 60 mph and get 30 mpg, then fuel consumption is:

$$\frac{dy}{dx} = 30 \times 60 = 1800 \text{ gallons per hour? Wait...}$$

Actually: $\frac{\text{miles}}{\text{gallon}} \times \frac{\text{miles}}{\text{hour}}$ doesn't work! Need consistent units.

Basic Chain Rule Examples

Example

Find the derivative of $y = (3x^2 + 1)^4$

Solution

1. Identify: Outer function: $(\ )^4$, Inner function: $3x^2 + 1$
2. Outer derivative: $4(\ )^3$
3. Inner derivative: $6x$
4. Apply chain rule: $y' = 4(3x^2 + 1)^3 \cdot 6x = 24x(3x^2 + 1)^3$

Example

Find the derivative of $y = \sqrt{x^3 - 2x}$

Solution

1. $y = (x^3 - 2x)^{1/2}$
2. Outer: $\frac{1}{2}(\ )^{-1/2}$, Inner: $3x^2 - 2$
3. $y' = \frac{1}{2}(x^3 - 2x)^{-1/2} \cdot (3x^2 - 2) = \frac{3x^2 - 2}{2\sqrt{x^3 - 2x}}$

Practice 1: Basic Chain Rule

Find the derivative of each function:

1. $y = (2x - 5)^3$
2. $y = \sqrt{4x^2 + 1}$
3. $y = \frac{1}{(x^2+3)^2}$
4. $y = (x^3 - 2x^2 + 1)^4$

Strategy

1. Identify inner and outer functions
2. Find derivative of outer function (keep inner unchanged)
3. Multiply by derivative of inner function
4. Simplify

Practice 1 Solutions

1. $y = (2x-5)^3$
   $y' = 3(2x-5)^2 \cdot 2 = 6(2x-5)^2$
2. $y = \sqrt{4x^2+1} = (4x^2+1)^{1/2}$
   $y' = \frac{1}{2}(4x^2+1)^{-1/2} \cdot 8x = \frac{4x}{\sqrt{4x^2+1}}$
3. $y = \frac{1}{(x^2+3)^2} = (x^2+3)^{-2}$
   $y' = -2(x^2+3)^{-3} \cdot 2x = -\frac{4x}{(x^2+3)^3}$
4. $y = (x^3-2x^2+1)^4$
   $y' = 4(x^3-2x^2+1)^3 \cdot (3x^2-4x) = 4(3x^2-4x)(x^3-2x^2+1)^3$

Multiple Compositions: Triple Chain Rule

Theorem (Extended Chain Rule)

If $y = f(g(h(x)))$, *then:*

$$\frac{dy}{dx} = f'(g(h(x))) \cdot g'(h(x)) \cdot h'(x)$$

Example

Find the derivative of $y = \sin^3(2x^2 + 1)$

Solution

1. $y = [\sin(2x^2 + 1)]^3$ (three layers!)
2. Outer: $3[\ ]^2$, Middle: $\cos(\ )$, Inner: $4x$
3. $y' = 3[\sin(2x^2 + 1)]^2 \cdot \cos(2x^2 + 1) \cdot 4x$
4. $y' = 12x \sin^2(2x^2 + 1) \cos(2x^2 + 1)$

Alternative Form: Generalized Power Rule

Theorem (Generalized Power Rule)

If $y = [u(x)]^n$, *then:*

$$\frac{dy}{dx} = n[u(x)]^{n-1} \cdot u'(x)$$

Example

1. $y = (x^2 + 3x)^5$
   $y' = 5(x^2 + 3x)^4 \cdot (2x + 3)$
2. $y = \sqrt[3]{2x - 1} = (2x - 1)^{1/3}$
   $y' = \frac{1}{3}(2x - 1)^{-2/3} \cdot 2 = \frac{2}{3\sqrt[3]{(2x-1)^2}}$
3. $y = \frac{1}{\sqrt{x^2+4}} = (x^2 + 4)^{-1/2}$
   $y' = -\frac{1}{2}(x^2 + 4)^{-3/2} \cdot 2x = -\frac{x}{(x^2+4)^{3/2}}$

Practice 2: Multiple Compositions

Find the derivative of each function:

1. $y = \cos(3x^2 - 2x)$
2. $y = \sqrt{\sin x + 1}$
3. $y = (2x + \sqrt{x^2 + 1})^3$
4. $y = \tan^4(2x)$

Tips

- Work from outside in
- For multiple compositions, apply chain rule repeatedly
- Check if you can simplify before differentiating

Practice 2 Solutions

1. $y = \cos(3x^2 - 2x)$
   $y' = -\sin(3x^2 - 2x) \cdot (6x - 2) = -(6x - 2)\sin(3x^2 - 2x)$
2. $y = \sqrt{\sin x + 1} = (\sin x + 1)^{1/2}$
   $y' = \frac{1}{2}(\sin x + 1)^{-1/2} \cdot \cos x = \frac{\cos x}{2\sqrt{\sin x + 1}}$
3. $y = (2x + \sqrt{x^2 + 1})^3$
   $y' = 3(2x + \sqrt{x^2 + 1})^2 \cdot \left(2 + \frac{x}{\sqrt{x^2+1}}\right)$
4. $y = \tan^4(2x) = [\tan(2x)]^4$
   $y' = 4[\tan(2x)]^3 \cdot \sec^2(2x) \cdot 2 = 8\tan^3(2x)\sec^2(2x)$

Generalized Derivative Formulas

With chain rule, we can extend all basic derivative formulas:
If $u = u(x)$ is differentiable:

- $\frac{d}{dx}[u^n] = nu^{n-1}u'$
- $\frac{d}{dx}[\sin u] = (\cos u)u'$
- $\frac{d}{dx}[\cos u] = (-\sin u)u'$
- $\frac{d}{dx}[e^u] = e^u u'$
- $\frac{d}{dx}[\ln u] = \frac{1}{u}u'$
- $\frac{d}{dx}[\tan u] = (\sec^2 u)u'$

For example:

1. $\frac{d}{dx}[\sin(x^2)] = \cos(x^2) \cdot 2x$
2. $\frac{d}{dx}[e^{3x}] = e^{3x} \cdot 3$
3. $\frac{d}{dx}[\ln(2x+1)] = \frac{1}{2x+1} \cdot 2$

Practice 3: Mixed Problems

Find the derivative of each function:

1. $y = e^{x^2+3x}$
2. $y = \ln(2x^3 - x)$
3. $y = \sin(2x)\cos(3x)$ (Product Rule + Chain Rule!)
4. $y = \frac{e^{2x}}{x^2+1}$ (Quotient Rule + Chain Rule!)

Advanced Strategy

- Identify all rules needed
- Apply product/quotient rule first if needed
- Then apply chain rule to individual parts
- Simplify step by step

Practice 3 Solutions

1. $y = e^{x^2+3x}$
   $y' = e^{x^2+3x} \cdot (2x+3)$
2. $y = \ln(2x^3 - x)$
   $y' = \frac{1}{2x^3-x} \cdot (6x^2 - 1) = \frac{6x^2-1}{2x^3-x}$
3. $y = \sin(2x)\cos(3x)$
   $y' = [\cos(2x) \cdot 2]\cos(3x) + \sin(2x)[-\sin(3x) \cdot 3]$
   $y' = 2\cos(2x)\cos(3x) - 3\sin(2x)\sin(3x)$
4. $y = \frac{e^{2x}}{x^2+1}$
   $y' = \frac{[e^{2x} \cdot 2](x^2+1) - e^{2x}[2x]}{(x^2+1)^2} = \frac{2e^{2x}(x^2+1) - 2xe^{2x}}{(x^2+1)^2}$
   $y' = \frac{2e^{2x}(x^2-x+1)}{(x^2+1)^2}$

Real-World Application: Related Rates

Example

A spherical balloon is inflated at 10 cm^3/s. How fast is the radius increasing when radius is 5 cm?

Solution

- Volume: $V = \frac{4}{3}\pi r^3$
- Differentiate with respect to time: $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$
- Given: $\frac{dV}{dt} = 10$, $r = 5$
- $10 = 4\pi(5)^2 \frac{dr}{dt}$
- $\frac{dr}{dt} = \frac{10}{100\pi} = \frac{1}{10\pi}$ cm/s

Chain Rule Connection

$\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$ is essentially the chain rule!

Common Mistakes to Avoid

Don't Forget the Inner Derivative!

- Wrong: $\frac{d}{dx}[(3x+1)^2] = 2(3x+1)$
- Correct: $\frac{d}{dx}[(3x+1)^2] = 2(3x+1) \cdot 3 = 6(3x+1)$

Don't Distribute the Power

- Wrong: $(x^2+1)^3 = x^6 + 1$ then differentiate
- Correct: Use chain rule directly

Order Matters in Multiple Compositions

- Work from outside in systematically
- Don't jump steps in complex compositions

Summary: Chain Rule Essentials

Key Formulas

- Basic: $(f \circ g)'(x) = f'(g(x)) \cdot g'(x)$
- Leibniz: $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$
- Generalized Power: $\frac{d}{dx}[u^n] = nu^{n-1}u'$

Strategy

1. Identify inner and outer functions
2. Differentiate outer function (keep inner unchanged)
3. Multiply by derivative of inner function
4. For multiple compositions, repeat the process

Applications

- Related rates problems
- Motion along curved paths
- Exponential growth/decay models
- Any situation with changing rates of changing quantities

Final Practice Problems

Find the derivative of each function:

1. $y = (x^3 - 2x + 1)^{10}$
2. $y = \sqrt[4]{3x^2 - 5x}$
3. $y = \cos(2x^3 - x)$
4. $y = e^{\sin x}$
5. $y = \ln(x^2 + 4x + 4)$
6. $y = (2x - 1)\sqrt{x^2 + 1}$ (Product + Chain)
7. $y = \frac{\sin(2x)}{e^x}$ (Quotient + Chain)

Challenge

Try to solve these without looking at previous examples first!